

EXPERIMENT-5a

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UI Design Lifecycle Using the RAD Model

User Interface (UI) design focuses on creating interactive systems that allow users to efficiently interact with software applications. A well-designed UI improves usability, accessibility, and user satisfaction. To develop effective interfaces quickly, the Rapid Application Development (RAD) model is often used.

The RAD model emphasizes fast prototyping, iterative development, and continuous user feedback. Instead of designing the entire system at once, developers build prototypes early and refine them through repeated testing and improvement.

RAD is particularly useful in UI design because visual interfaces can be easily modified based on user feedback during the development process. The model consists of four major phases: Requirements Planning, User Design, Construction, and Cutover.

Phase 1: Requirements Planning

The first phase focuses on identifying the basic requirements of the system. Designers determine what features the user interface should provide and how users will interact with the application.

During this stage, the development team identifies key components such as navigation structure, important screens, and user actions. Clear requirements help guide the later stages of interface design.

For the Shopping Application prototype, the key requirements include:

- Navigation: Home, Products, Cart, Orders, Profile
- User Actions: Browse products, view details, add to cart, checkout, track orders
- Display available products and categories
- Allow users to search and filter items
- Show cart summary and total price
- Enable secure payment and order confirmation
- Track order status and delivery history

Phase 2: User Design

In the User Design phase, wireframes and interactive prototypes are created to represent the layout and flow of the application. Designers plan every screen, defining buttons, navigation, input fields, and visual hierarchy.

Tools Used

- Axure RP – for interactive wireframing and prototyping
- Figma / Canva – for visual UI design

Designed Screens

- Home Page – Displays featured products and categories
- Product Listing Page – Shows grid/list of available items
- Product Detail Page – Full product info with add-to-cart button

- Cart Page – Lists selected items with quantity and price
- Checkout Page – Delivery address and payment options
- Order Confirmation Page – Summary and confirmation message

Design Features

- Simple and user-friendly interface
- Clear navigation between screens
- Reusable components like header, footer, and product cards
- Buttons for search, add to cart, and checkout

The following screenshots show the UI screens designed during this phase:

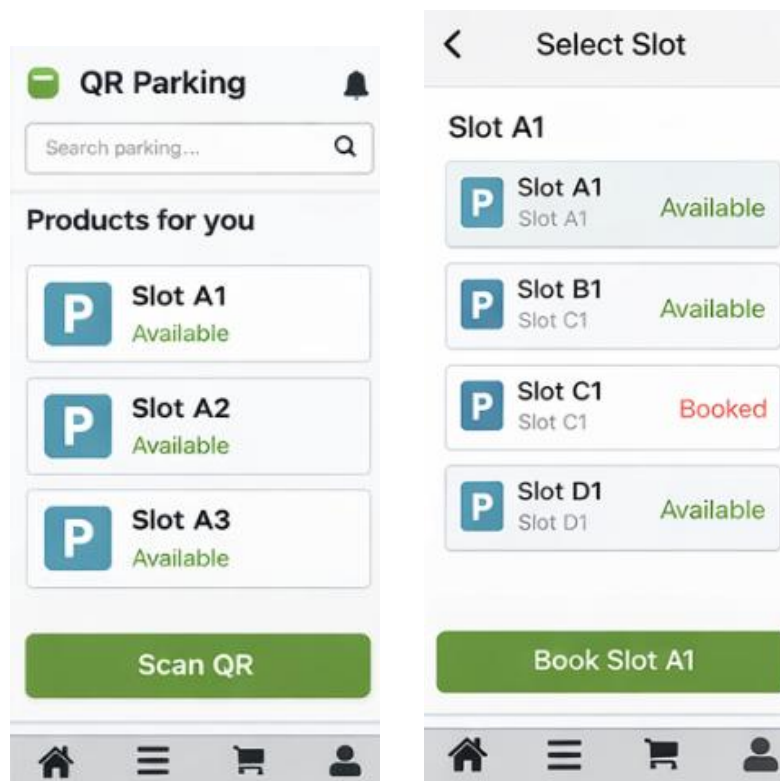


Fig 1: Home Page (left) | Product Listing Page (right)

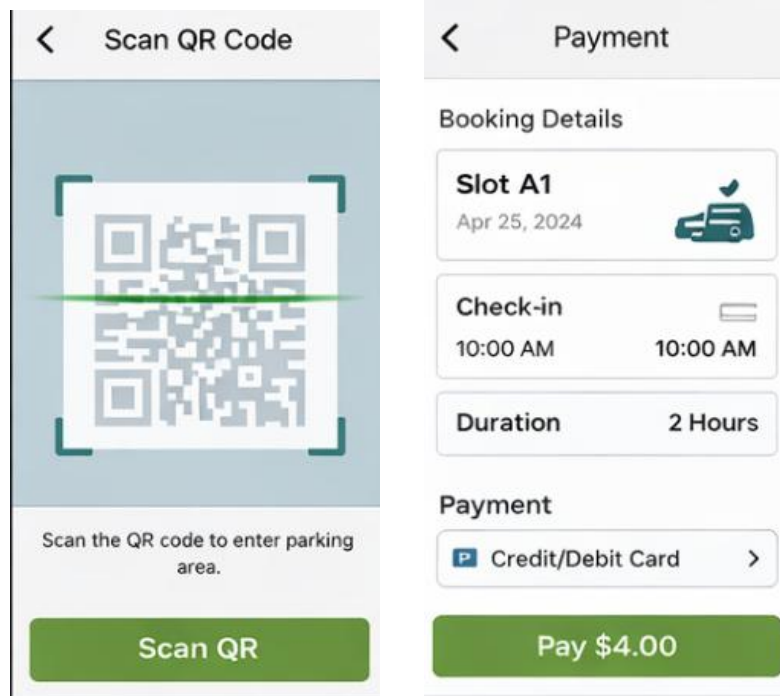


Fig 2: Product Detail Page (left) | Cart Page (right)

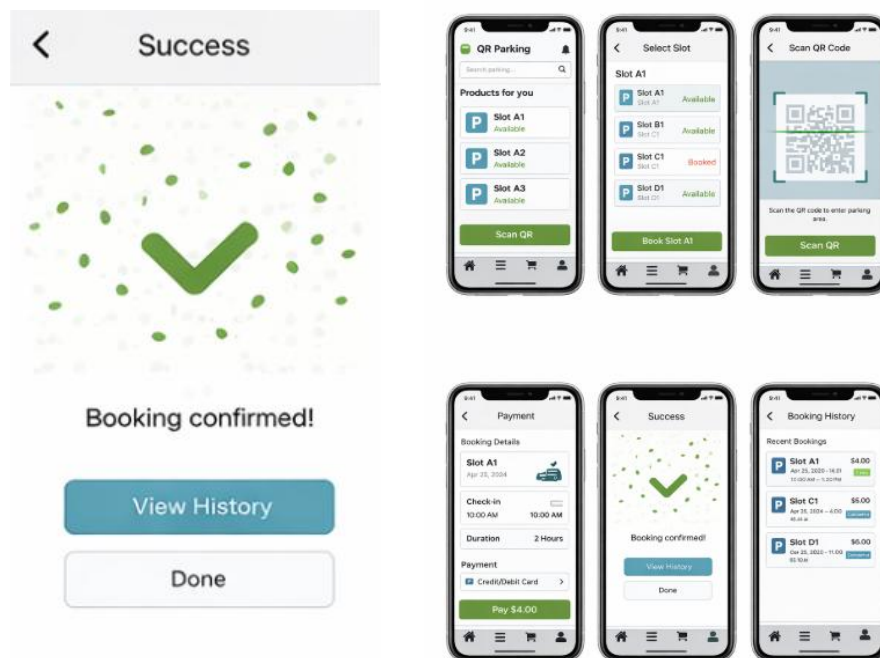


Fig 3: Checkout Page (left) | Order Confirmation Page (right)

Phase 3: Construction

The Construction phase involves converting the designed wireframes into a fully interactive prototype. Navigation between screens is implemented, and interactive behaviors such as button clicks, transitions, and dynamic elements are added.

Prototype Flow

Home → Product Listing → Product Detail → Cart → Checkout → Payment → Order Confirmation

Dynamic Features

- Product search and category filtering
- Add to cart with quantity update
- Cart summary with live price calculation
- Checkout form with address input
- Payment processing simulation screen
- Order confirmation with tracking ID

Testing

- Prototype tested using preview mode in Axure RP
- Feedback collected from users on navigation and layout
- UI improved based on suggestions (button placement, font size, color scheme)
- Multiple iterations performed until usability goals were met

Phase 4: Cutover

The Cutover phase represents the final stage of the RAD lifecycle. In this stage, the completed interface prototype is finalized and prepared for deployment or handoff to the development team.

Finalization

- Prototype shared via Axure Cloud link for stakeholder review
- All screens validated for consistent design and navigation
- Ready for front-end development handoff
- Can be converted into a real mobile or web application

User Support

- User instructions and walkthrough guide provided
- Easy navigation ensured for all types of users
- Documentation shared with the development team

Role of Axure RP in UI Prototyping

Axure RP is a professional prototyping tool used for designing interactive user interfaces. It provides a drag-and-drop environment where designers can quickly build wireframes and simulate application behavior.

Key features of Axure RP include:

- Creating wireframes and screen layouts

- Adding interactive elements such as buttons, menus, and input fields
- Defining navigation between screens with conditional logic
- Building reusable components using masters and dynamic panels
- Exporting interactive prototypes for testing and sharing

Using Axure RP, designers can quickly develop and demonstrate UI concepts without writing actual code. The tool bridges the gap between design and development by producing realistic, clickable prototypes.

Conclusion

The RAD model provides an efficient and user-centered approach to UI design. By emphasizing rapid prototyping, iterative improvement, and continuous user feedback, the model enables designers to create high-quality interfaces in less time.

The Shopping Application prototype developed through this process demonstrates how each RAD phase contributes to a complete and polished UI design. The use of tools like Axure RP further accelerates the design cycle, making it suitable for modern application development.